

## Hands-On Lab | Exercise 1

HTML + CSS: Read It, Break It, Fix It, Style It

USTH Web Application Development 4 hours VS Code + Chrome

### Learning Outcomes

By the end of this session, you will be able to:

- Read a piece of HTML and CSS and **predict** what the browser will draw, before running it.
- Explain which rule wins when several CSS rules hit the same element.
- Compute the width a box takes on screen from its `width`, `padding`, `border` and `margin`.
- Find a mistake in someone else's HTML or CSS and describe its **symptom**, its cause, and its fix.
- Inspect any live web page with DevTools and read the Styles and Computed panels.
- Style a page you are not allowed to edit, using selectors alone.

### Setup

**Editor:** VS Code with the **Live Server** extension (free).

**Browser:** Chrome or Firefox - you will need `F12`.

**Files from your instructor:** `broken.html` (Part 2) and `ict-department.html` (Part 4).

**Paper:** Parts 1 and 2 start on paper with the laptop closed. Bring a pen of a second colour.

### ⚠ How this exercise is marked

You may use AI tools. The marks are not in the code - they are in the boxes where you write **what you predicted**, **what you observed**, and **why the two differ**.

Every part has two stages:

| Stage                  | What you do                                                        | Tools allowed |
|------------------------|--------------------------------------------------------------------|---------------|
| <b>A - On your own</b> | Write your answer on paper. Browser closed, AI closed.             | none          |
| <b>B - Check</b>       | Run it, open DevTools, ask AI. Correct Stage A in a second colour. | everything    |

**Hand in Stage A before you start Stage B.** A sheet with an empty Stage A scores zero, even if Stage B is perfect. At the end of the session your teacher will point at one line of your own work and ask you to explain it out loud.

| Part | Topic                                         | Time   |
|------|-----------------------------------------------|--------|
| 1    | Read the code, predict the screen             | 45 min |
| 2    | Find the mistakes                             | 45 min |
| 3    | Look inside a real page with <code>F12</code> | 45 min |
| 4    | Style a page without touching the HTML        | 90 min |
|      | Hand in + oral check                          | 15 min |

## Part 1 Read the Code, Predict the Screen (45 min)

Each question gives you the code and a **Toolbox** listing exactly which tags and which CSS properties are in play. The Toolbox tells you what to revise; it does not give the answer.

Write your prediction on paper. *Then* type the code into VS Code and compare.

### Q1 How many lines appear on screen?

```
<div>Pho bo</div>
<div>Pho ga</div>
<span>Bun cha</span>
<span>Banh mi</span>
<p>Com tam</p>
```

#### Toolbox

**Tags:** `<div>` `<span>` `<p>`

**Concept:** block-level vs inline

**Q1.** How many separate lines does the browser draw? \_\_\_\_\_

**Q2.** Write down which text sits on which line.

**Q3.** `<span>` and `<div>` have no colour and no border of their own. So why does the screen look different?

### Q2 Which colour wins?

```
<p id="first" class="note highlight">Beef noodle soup</p>

p      { color: gray; }
.note  { color: blue; }
.highlight { color: green; }
#first { color: red; }
```

#### Toolbox

**Tags:** `<p>` **Attributes:** `id` `class`

**CSS:** `color` **Selectors:** type, class `.`, `id` `#`

**Concept:** specificity, order of rules

**Q1.** What colour is the text? \_\_\_\_\_

**Q2.** Delete the `#first` rule. What colour now? \_\_\_\_\_

**Q3.** Also swap the `.note` and `.highlight` lines. What colour now? \_\_\_\_\_

**Q4.** The element carries *two* classes. Do two classes beat one id? Answer yes or no, and say why.

### Q3 How much space does the box take?

```
<div class="card">Pho Thin</div>

.card {
  width: 300px;
  padding: 15px;
  border: 5px solid black;
  margin: 10px;
}
```

 **Toolbox****Tags:** `<div>`**CSS:** `width padding border margin`**Concept:** the box model - content, padding, border, margin

| Measurement                               | px |
|-------------------------------------------|----|
| content width                             |    |
| padding left + right                      |    |
| border left + right                       |    |
| <b>width of the box the browser draws</b> |    |
| margin left + right                       |    |
| <b>total horizontal space used</b>        |    |

**Q1.** Which of the four boxes takes the background colour, and which one never does?

**Q4 Same CSS, one extra line**

```
.card {
  width: 300px;
  padding: 15px;
  border: 5px solid black;
  margin: 10px;
  box-sizing: border-box;
}
```

 **Toolbox****CSS:** `box-sizing` - values `content-box` (the default) and `border-box`**Concept:** what the number in `width` actually measures

**Q1.** The box the browser draws is now how many px wide? \_\_\_\_\_

**Q2.** The content inside it is now how many px wide? \_\_\_\_\_

**Q3.** In one sentence: with `border-box`, what does the number in `width` mean?

## Q5 The style attribute

```
<p class="price" style="color: black;">60,000 VND</p>

.price { color: red; }
p      { color: blue; }
```

### Toolbox

**Tags:** `<p>` **Attributes:** `class` `style`

**CSS:** `color`

**Concept:** inline style vs stylesheet

**Q1.** What colour is the price? \_\_\_\_\_

**Q2.** You are *not* allowed to edit the HTML. Write a CSS rule that turns the price red anyway.

**Q3.** Your teacher tells you never to write that rule in real work. Why not?

## Part 2 Find the Mistakes (45 min)

The file below is supposed to show the Pho Thin menu page. It opens in the browser without any error message, but **nine things are wrong**. None of them is a typo in the *text*. Some are things written wrongly; at least one is a line that should be there and is not.

### What to do

**Stage A - on paper, laptop closed.** Fill in the table. The **Symptom** column is the important one: describe what a person *looking at the screen* would see, before you say how to fix it.

**Stage B - check.** Open the file in VS Code, press `F12`, correct yourself in a second colour. Ask AI *only after* Stage A is handed in; note anything it caught that you missed.

### broken.html

```

1 <!DOCTYPE html>
2 <html>
3 <head>
4   <title>Pho Thin - Menu</title>
5   <style>
6     body { font-family: Arial, sans-serif; }
7     h1 { color: #b00000; text-align: center; }
8     title { font-style: italic; color: gray; }
9     .price { font-weight: bold
10              color: #b00000; }
11     table, th, td { border: 1px solid #999; padding: 8px; }
12     th { background-color: #b00000; color: white; }
13   </style>
14 </head>
15 <body>
16   <h1>Pho Thin</h2>
17   <p class="title">Since 1979</p>
18
19   <h2>Our bowls</h2>
20   <ul>
21     <li>Pho bo</li>
22     <li>Pho ga</li>
23   </ul>
24   <li>Pho tai</li>
25
26   <h2>Prices</h2>
27   <table>
28     <tr><th colspan="3">Menu of the day</th></tr>
29     <tr><td>Pho bo</td><td class="price">60,000</td></tr>
30     <tr><td>Pho ga</td><td class="price">55,000</td></tr>
31   </table>
32   <h2>Find us</h2>
33   <p>
34     <a href="https://phothin.vn">Visit our website
35     Open 6am - 10am, closed on Monday.
36   </p>
37
38   
39 </body>
40 </html>

```

**Answer table**

| # | Line | Symptom on screen | Cause | Fix |
|---|------|-------------------|-------|-----|
| 1 |      |                   |       |     |
| 2 |      |                   |       |     |
| 3 |      |                   |       |     |
| 4 |      |                   |       |     |
| 5 |      |                   |       |     |
| 6 |      |                   |       |     |
| 7 |      |                   |       |     |
| 8 |      |                   |       |     |
| 9 |      |                   |       |     |

**Stage B - what AI caught that you did not:****Three questions to answer under the table**

- Q1.** In two places the CSS contains a **value the browser cannot understand**, so it throws that declaration away and says nothing at all. Which two? Explain how the browser decides to discard a declaration instead of complaining. *Careful with one of them - more is lost than you first think.*
- Q2.** One mistake makes text that is *not* inside a link behave like a link. Which one, and why does the damage spread past the line where the mistake is?
- Q3.** One mistake costs nothing today because the image file is missing anyway, but it would still be wrong if the image loaded. Which one, and who is harmed by it?

## Part 3 Look Inside a Real Page with F12 (45 min)

Every page you have ever visited hands its HTML and CSS to your browser. DevTools is how you read it. This is the panel you will use for the rest of your career when a page does not look the way you expect.

### 3.1 The five things you need to know

1. **Open it.** Press `F12`, or `Ctrl+Shift+I`, or right-click anything on the page and choose **Inspect**. On a Mac: `Cmd+Option+I`.
2. **Find the Elements tab.** Leftmost tab. The left half shows the page as a tree of tags. Hover a line in the tree and the matching part of the page lights up.
3. **Use the picker.** The arrow-in-a-box icon at the top left, or `Ctrl+Shift+C`. Click it, then click anything on the page. DevTools jumps straight to that element.
4. **Read the Styles pane.** The right half. It lists every CSS rule that hits the selected element, **strongest first**.
5. **Read the Computed tab.** Next to Styles. This is the **final answer** - the value the browser actually used after all the rules fought it out.

#### 💡 Styles vs Computed

**Styles** shows you the *argument*. **Computed** shows you the *verdict*.

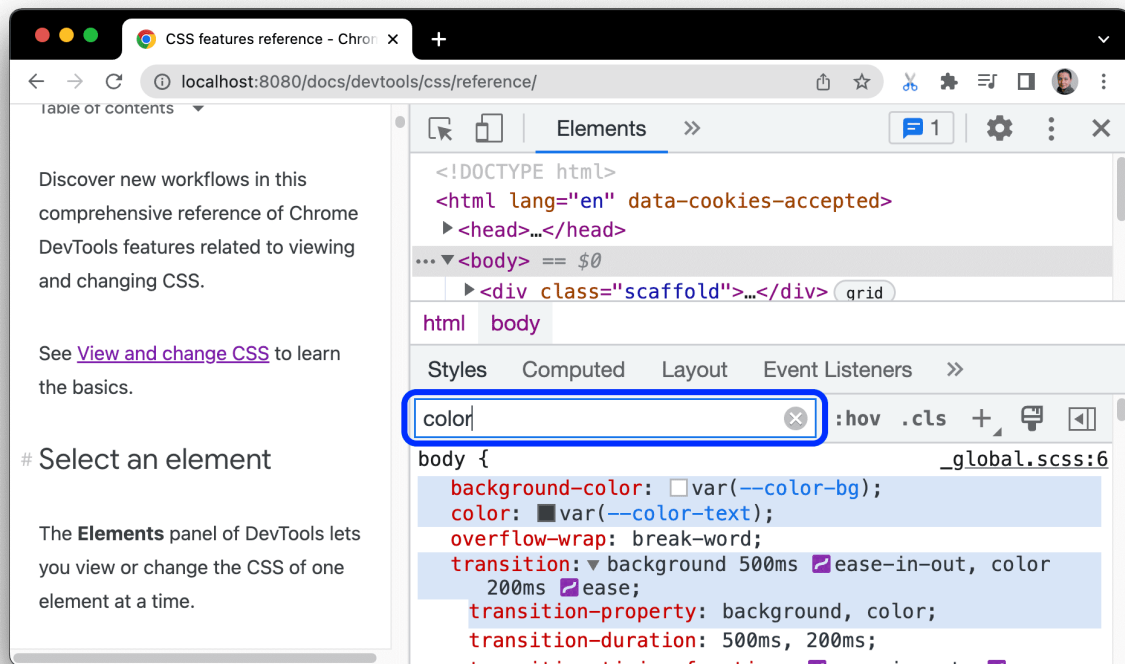
Three things to notice in the Styles pane:

- A declaration with a **line through it** lost - another rule beat it, or the value is invalid.
- `element.style { }` at the very top means an inline `style="..."` attribute on the tag.
- The file name and line number on the right of each rule tell you which stylesheet it came from.

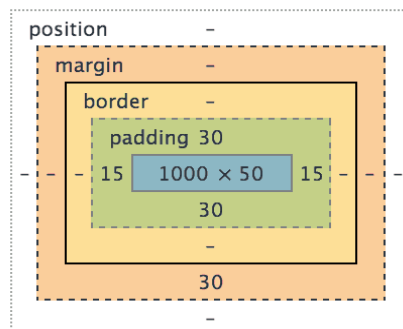
Scroll to the bottom of **Computed** and you will find the box-model diagram: four coloured rings showing content, padding, border and margin in pixels.

#### 🖱️ Two more things you will use constantly

- Each declaration in Styles has a **checkbox** when you hover it. Untick it to switch that one declaration off and watch the page change. Tick it back on.
- Click any **value** in Styles and type a new one. The page updates as you type.



Styles    Computed    Event Listeners    DOM Breakpoints    Properties



☐ Show inherited properties

- webkit-background-clip: border-box;
- webkit-background-origin: padding-box;
- webkit-background-size: auto;
- ▶ -webkit-box-shadow: rgba(0, 0, 0, 0.0980392) 0px 1px 0px 0px;
- ▶ background-attachment: scroll;
- ▶ background-clip: border-box;
- ▶ background-color: rgb(255, 255, 255);
- ▶ background-image: none;
- ▶ background-origin: padding-box;
- ▶ background-size: auto;

### 3.2 Investigate a real site

Open <https://usth.edu.vn> in Chrome or Firefox. If the site has changed and a question no longer makes sense, say so in your answer and use the closest equivalent element.

#### Task 1 - Identify

Use the picker on the **main heading of the home page**.

| Question                                       | Your answer |
|------------------------------------------------|-------------|
| Which tag is it?                               |             |
| Does it carry a <code>class</code> ? Write it. |             |
| Does it carry an <code>id</code> ? Write it.   |             |

#### Task 2 - Declared vs computed

Still on that heading:

| Question                                                            | Your answer |
|---------------------------------------------------------------------|-------------|
| In <b>Styles</b> , which <code>font-family</code> list is declared? |             |
| In <b>Computed</b> , which single font is actually used?            |             |
| Why are those two answers different?                                |             |
| The text colour in <b>Computed</b> , as <code>rgb(...)</code>       |             |
| The same colour written as hex                                      |             |

#### Task 3 - The box

Scroll to the box-model diagram at the bottom of Computed, for the same heading.

|         | top | right | bottom | left |
|---------|-----|-------|--------|------|
| padding |     |       |        |      |
| border  |     |       |        |      |
| margin  |     |       |        |      |

Content width \_\_\_\_\_px. Total horizontal space used \_\_\_\_\_px. Show your arithmetic:

#### Task 4 - Who won the fight

Still on the heading, look at the whole Styles pane.

**Q1.** How many separate rules target this element? \_\_\_\_\_

**Q2.** Find one declaration with a line through it. Write the selector it belongs to. \_\_\_\_\_

**Q3.** Write the selector of the rule that **beat** it. \_\_\_\_\_

**Q4.** Explain in one sentence why the winner won - was it specificity, or was it order?

**Task 5 - Switch things off**

1. Find any element with a `background-color` . Untick that declaration. Screenshot the before and the after.
2. Find an element with `display: flex` . Change the value to `block` . Describe in two sentences what moved and why.
3. Set the `<body>` `font-family` to `monospace` . Screenshot.

**Task 6 - The question that matters**

You have just recoloured and rearranged the USTH home page.

- Q1.** Press `F5` . What happened to your changes?
- Q2.** Ask a classmate to load the same page. Do they see your changes?
- Q3.** Explain, using what you learned in Lecture 1 about client and server: **where** did your edits live, and why did the server never know about them?

**Task 7 - Two ways of looking**

Press `Ctrl+U` on the same page. This is **View Source**.

- Q1.** Copy the first five lines you see.
- Q2.** Compare with the Elements tab. Are they the same? Find one thing that appears in Elements but not in View Source.
- Q3.** Explain the difference in one sentence. Which one is the file the server sent, and which one is what the browser has built in memory?

 **What to hand in for Part 3**

One document containing the four tables filled in, the four screenshots, and your written answers to Tasks 4, 6 and 7.

## Part 4 Style a Page Without Touching the HTML (90 min)

`ict-department.html` is a full page: banner, navigation, two columns, cards, two tables, a news list and a footer. It already carries every `class` and `id` you will need. It has no styling at all.

### 🚫 The one rule

**You may not change a single character of the HTML file.** You write `style.css`. Nothing else. The stylesheet is already linked on line 6 of the HTML - create the file next to it and start writing.

Every task has a **Why** question. The CSS is worth half the marks; the **Why** is worth the other half, and your teacher will pick two at random and ask you out loud.

### 4.1 `ict-department.html`

```

<!DOCTYPE html>
<html>
<head>
  <meta charset="utf-8">
  <title>ICT Department - USTH</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>

<!-- ===== BANNER ===== -->
<header class="banner">
  <h1 class="site-name">Information and Communication Technology</h1>
  <p class="tagline">University of Science and Technology of Hanoi</p>
</header>

<!-- ===== NAVIGATION ===== -->
<nav class="main-nav">
  <a href="#about" class="nav-link">About</a>
  <a href="#programs" class="nav-link">Programs</a>
  <a href="#timetable" class="nav-link">Timetable</a>
  <a href="#credits" class="nav-link">Credits</a>
  <a href="#news" class="nav-link current">News</a>
</nav>

<!-- ===== TWO COLUMNS ===== -->
<div class="layout">

  <!-- ---- LEFT COLUMN ---- -->
  <aside class="sidebar">
    <h3 class="side-title">Quick links</h3>
    <ul class="side-list">
      <li><a href="#" class="side-link">Student portal</a></li>
      <li><a href="#" class="side-link">Library</a></li>
      <li><a href="#" class="side-link">Lab booking</a></li>
      <li><a href="#" class="side-link">Contact office</a></li>
    </ul>
    <p class="side-note" style="color: #888;">Office hours 8am - 5pm</p>
  </aside>

  <!-- ---- RIGHT COLUMN ---- -->
  <main class="content">

    <section id="about" class="block">
      <h2 class="block-title">About the department</h2>
      <p class="lead">The ICT department trains engineers in software, networks and data.</p>
      <p>Teaching is in English. The bachelor program follows the European Credit Transfer
        and Accumulation System, with 180 credits over three years.</p>
      <p class="lead">Students join a research lab in the third year.</p>
    </section>

    <section id="programs" class="block">
      <h2 class="block-title">Programs</h2>
      <div class="card-row">

```

```


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HTML + CSS


```

```
<p>18 Hoang Quoc Viet, Cau Giay, Hanoi & middot; officeusth@usth.edu.vn</p>
</footer>

</body>
</html>
```

## 4.2 The tasks

- T1. Banner.** Dark red background `#b00000`, white text, 24px of space inside on every side, text centred.  
*Why:* you needed both `color` and `background-color`. Explain what each one paints. Then explain why `padding: 24px` and not `margin: 24px` - what would `margin` have given you instead?
- T2. Navigation bar.** The five links sit in one horizontal row, 20px apart, starting from the left. No underlines. Dark red on hover.  
*Why:* (a) which element did you put `display: flex` on, and why not on `.nav-link`? (b) You used `gap`. What would you have written instead if `gap` did not exist? (c) `a:hover` is a pseudo-class - in one sentence, what does that mean?
- T3. The current page marker.** The link with `class="nav-link current"` is bold and dark red *at all times*, not only on hover.  
*Why:* that element carries two classes. Explain which rules apply to it, and what would happen if you had written `.current` above `.nav-link` in your file instead of below.
- T4. Two columns.** The sidebar is exactly 220px wide and must not shrink. The main content takes all the remaining width. 30px between the two columns.  
*Why:* explain which element became the flex container and which two became the flex items. Then explain what happens if you give the sidebar `width: 220px` and nothing else - why might it still shrink?
- T5. Sidebar links.** Remove the bullet points from `.side-list`. Each `.side-link` is dark grey, no underline, dark red on hover.  
*Why:* you removed the bullets with a property on the *list*, not on the items. Explain why that works.
- T6. The stubborn note.** `.side-note` must be dark red.  
*Why:* this one is the trap. Write down what you tried first, what happened, and what you had to write in the end. Then answer: given that you cannot edit the HTML, is your solution something you would write in a real project? Explain.
- T7. Lead paragraphs.** Inside `#about`, the paragraphs with `class="lead"` are 18px and dark red. The plain paragraph stays as it is.  
*Why:* there are three `<p>` in that section. Explain how the browser knows which two to style, and what would have happened if you had used `#about p` as your selector instead.
- T8. The cards.** The three cards sit in one row, share the width equally, have 16px of padding, a 1px grey border, and 20px between them.  
*Why:* if you set each card to `width: 33%` and give them padding, the third card drops to the next line. Explain why, and give **two** different ways to prevent it. Say which one you chose and why.
- T9. The badge.** `.badge` is a small dark red label with white text and 4px of padding. The one that also has `class="badge closed"` is grey instead.  
*Why:* `<span>` is inline. Explain what that means for `padding-top` on a badge, and what you would change if you wanted the badge to control its own height.

**T10. The timetable.** Single lines, not doubled. 8px of padding in every cell. Header cells dark red with white text. Empty cells must still show their border.

*Why:* (a) which declaration merged the doubled lines, and which element did it go on? (b) An empty `<td>` sometimes loses its border - explain what causes that and how your CSS avoids it.

**T11. The credits table, and only the credits table.** `#ects-table` gets a dark blue header, a light grey background on the whole table, and the row under the mouse turns pale yellow. **The timetable above must not change.**

*Why:* you had a choice between `.credits` and `#ects-table`. Both work here. Explain what each one would hit if a second credits table were added to the page later, and which one you would choose in a real project.

**T12. News list.** No bullets. Each `.news-date` is grey, monospace, and 70px wide so all three dates line up in a column. Each `.news-link` has no underline and turns dark red on hover.

*Why:* `<span>` is inline, and `width` does nothing on an inline element. Explain what you had to do to make the 70px take effect, and what that change did to the layout of the line.

**T13. Footer.** Dark red background, white text, centred, 16px of padding.

*Why:* the footer background stretches across the full page width without you setting a width.

#### A note on AI

Nothing here forbids you from asking an AI. But an AI can hand you working CSS in five seconds; it cannot tell your teacher why you chose `border-box` over `width: 30%`, and it cannot look at the broken layout on your own screen. The point of the session is that when you do use an AI - next week, at work - you can tell in ten seconds whether what it gave you is right.